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Full Stack Development / Further Web Programming
COSC2758 UG /COSC2938 PG (Semester 1, 2026)
Assignment 1

Initial published version 13/03/2026 - please review and report any typo, broken link, inconsistency etc. This document will be considered final on the 16/03/2026

Assessment Type	This is a pair programming assignment which is to be attempted in a group of 2. • NO INDIVUAL SUBMISSIONS ALLOWED, unless you have a pre-approved Special Consideration. • Mixing a group with Undergraduate + Postgraduate students is not allowed. • You need to register the group on Canvas -> People -> Group, then add yourself into one of the pre-registered groups, according to the class that you are officially enrolled in. If you cannot find a suitable partner from your current class, you may join someone from another class, but you need to ensure that you chase for feedback from the tutor of that class, instead of from 2 tutors.
	Submit online via Canvas → Assignments → Assignment 1. Marks awarded for meeting requirements as closely as possible. Please read rubrics for details. Clarifications/updates may be made via announcements/relevant discussion forums.
Due Date	Week 6, Sunday 19th April 2026, 11:59 pm (midnight) *Melbourne Time*
Marks	25

1.1 Overview & Learning outcomes

You will use **React with TypeScript** (**TS**) to create a ***client-side*** *prototype of the web application*.

React with JavaScript will fetch ZERO for assignment 1.

No database is to be used at this stage. Use of a database will fetch ZERO for assignment 1.

The server-side implementation and full-stack development of the web application from assignment 1 will be completed in assignment 2.

The tasks are divided into *four* parts: PA (Pass), CR (Credit), DI (Distinction) & HD (High Distinction).

> The DI & HD section tasks will require self-research; you will not get straight answers in the course material. While we are happy to assist you with those tasks, you must do most of the work and research. This is done to prepare you for your future work and the rigors of the technological industry.

If you find a specification open to interpretation, post a query identifying the specification in the corresponding discussion board for Assignment 1. Software development in real life does not come with a definitive roadmap and flowcharts complete with instructions. More often than so, it is the developer's job to clarify client requirements. For this assignment, your prac tutor is considered as the client.

This assessment relates to the following learning outcomes of the course:

- CLO2: Demonstrate proficiency with a web application development framework
- CLO3: Implement a range of techniques and procedures for developing a small to medium-scale web application
- CLO5: Design and manage the development life cycle of a complete application.

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1.2 Group Registration on Canvas and GitHub Classroom

1.2.1 Register yourself in group of maximum two members. Registration is by moving your name to one of the pre-listed groups on Canvas -> People -> Group. You must make note of your team's name and use the exact same name for your GitHub Classroom team.

- If you cannot find a partner from your prac class, please team up with someone from another class by allocating yourself into a prac group, but you must attend that class weekly and maintain regular communication and chase feedback from the same tutor.
- If you happen to work alone, please still register yourself to a group according to the class that you are currently enrolled in, this is to ensure that you will be assigned a tutor to give feedback and grade your work.

1.2.2 Create a GitHub Classroom team repo by clicking the following invitation link https://classroom.github.com/a/1ziuEB2b

- You must use this particular repo for this assignment. Creating your repo somewhere else will simply make your work ineligible for any mark.
- The first team member needs to register team name according to your team's name on Canvas such as FWP PRA01/03 Wed 6.30pm Prakash Team 1, the second team member needs to select the newly created team and join it.
- Please do not use any other name, because your tutor needs to be able to find you on GitHub Classroom teams and match them with the registered teams on Canvas to be able provide marks and feedback on Canvas.
- Please check if you are enrolled in FSD for undergrad or FWP for postgrad. Do not join a PG group if you are and undergrad, and do not go join UG group if you are a postgrad.
- A repo with incorrect names or do not match with any of the Canvas group names will simply be deleted, and you will need to repeat the process.

1.2.3 Once accepted you and your partner will be given a dedicated repo that you should keep private and only be accessible by the team and your class tutor. By default, this repo should be hosted on GitHub classroom which mean your tutor can already see it. Please check with your tutor if they can see your repo, because they will need to be able to grade your assignment in the end.

1.3 Submission

You will be marked on the use of GitHub and the development process besides on the design and implementation quality of the actual software product that you are developing.

Please submit a **single zipped archive** with the following name format studentId1_studentId2.zip containing

- Project files without node_modules (please delete node_modules folder, and / or ignore it when you push it on GitHub)
- Group Contribution Form and
- README.md file containing group details (name, student IDs, what prac class are you enrolled in from according to myTimetable) and URL of your GitHub repo.

Git hosting by itself does **not** qualify for a submission and will NOT BE MARKED. You MUST submit the above-zipped archive.

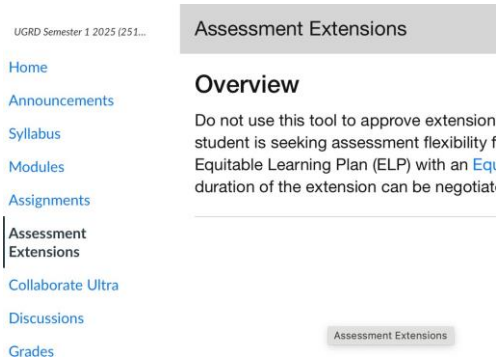
Since you will be submitting online, you agree to the Declaration stated below:

https://www.rmit.edu.au/students/my-course/assessment-results/assessment

1.4 Extension requests

After the due date, you will have **five** business days to submit your assignment as a late submission. Late submissions will incur a penalty of 10% per day. After these 5 days, Canvas will be closed, and you will lose ALL the assignment marks.

All extension requests must be made via the **Assessment Extensions** page of the Course canvas (*screenshot below*):



- We will **not** reply to email-related requests.
- Requests must be made at least **24 hours** before the assessment deadline.
- Only **one** of you should request on behalf of the group. MULTIPLE requests from the same group will be ignored.
- We can only offer you an extension of up to **7 days**. For longer extensions, please apply via the special consideration website: https://www.rmit.edu.au/students/my-course/assessment-results/special-consideration-extensions/special-consideration

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1.5 Academic integrity and plagiarism

Academic integrity is about the honest presentation of your academic work. It means acknowledging the work of others while developing your insights, knowledge and ideas. You should take extreme care that you have:

- acknowledged words, data, diagrams, models, frameworks and/or ideas of others you have quoted (i.e. directly copied), summarised, paraphrased, discussed or mentioned in your assessment through the appropriate referencing methods,
- provided a reference list of the online code used so your reader can locate the source if necessary. This includes material taken from Internet sites **& your own course material**.

If you fail to cite the sources of your material properly, you risk being accused of plagiarism for presenting someone else's work and ideas as your own without proper attribution.

RMIT University treats plagiarism as a very serious offence constituting misconduct. Plagiarism covers a variety of inappropriate behaviours, including:

- Contract cheating- paying someone to do your work
- Failure to properly document a source
- Copyright material from the internet or databases
- Collusion between students
- Posting assignment tasks on technical forums (*reddit, stack exchange, etc.*) and asking for solution(s)
- Copying code with the aid of generative AI tools.

For further information on our policies and procedures, please refer to:

https://www.rmit.edu.au/students/student-essentials/assessment-and-

results/academic-integrity

1.6 Use of generative AI tools

As we increasingly integrate technology into our academic and professional practices, it is imperative to address the use of generative AI tools, such as ChatGPT, in your university assessments. These tools should be used cautiously, as they function by synthesising and repackaging existing text from various sources. We will discuss the implications of using such technologies, including the academic standards and their significance in your future careers, during lectorial sessions. We encourage you to produce and submit only your original work and to seek clarification from instructors if you have any uncertainties.

Furthermore, please note that the Co-pilot feature will **not** be enabled within the official RMIT's Git organisation for this course (**rmit-fsd-2026-s1**). If you choose to use Co-pilot independently, it is crucial to use it responsibly. Remember, generative AI should not be used to replicate code directly.

We are committed to academic integrity and will conduct face-to-face meetings to address any cases of suspicious code submissions.

Ensure you properly reference any code that AI tools generate to maintain transparency and uphold academic honesty. Here are RMIT's official AI-related referencing guidelines:

https://rmit.libguides.com/referencing_AI_tools#s-lg-box-22377324

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1.7 Assessment scenario overview & system requirements

Your team has been selected by a events venue management company to develop Venue Vendors (VV), a web system dedicated to the selection and hiring of venues for events offered by Venue Vendors, a private company based in Melbourne that aims to finding and hiring venues quick and seamless.

Scope of assessment 1: involves developing the VV web system utilising ReactTS to create a responsive, interactive user interface *prototype for front-end development*. The system is designed to streamline the process of hiring venues by facilitating robust interactions among two key user groups: hirers and vendors. Vendors can choose and review hiring applicants, including leaving comments on profiles detailing the applicants' credibility. Conversely, applicants can fill out comprehensive application forms detailing their event details, hiring reputation, and compliant documents.

Out-of-scope *for the current assessment*

Later assessment 2, will involve full stack development and administrators managing the backend functionalities, such as populating form fields with relevant data and ensuring that information like course lists and departmental details are up to date. They will also be responsible for generating actionable reports that can help decision-making. The completed system will ensure data integrity and security to protect sensitive information. Additionally, it will support scalability to accommodate an increasing number of users and data over time.

Kindly note the following **10** system requirements:

a. The VV prototype user interface **must** have distinguishable areas: a header, a footer, main areas, and a navigation bar.

b. The home page **should** display relevant information for the applicants and the venues

c. The navigation bar **contains** the links: *sign-up, sign-in & sign-out* and anything else you deem relevant.

d. **Hiring applicants** submit their details.

e. A **Vendor** can choose the above applicants.

f. All the data will be stored in **HTML5's localStorage**. Using databases (MongoDB, Relational databases and/or Firebase) is **not** allowed.

g. Use of REST API of any kind is **not** allowed for this assessment.

h. The website **must** be fully styled and look professional. The content must make sense i.e., use of lorem ipsum is **not** allowed.

i. Only well-established and documented third-party libraries/modules **should** be used. If you are in doubt about a module's legitimacy, contact your tutor for advice.

j. The digital assets (images, icons, audio & video) must be outsourced from royalty-**free** websites. You should **not** steal someone else's assets to enhance the look and feel of your website. High-quality & free assets can be obtained from:

https://unsplash.com/ (images)
https://uifaces.co/ (avatars)
https://fonts.google.com/icons?selected=Material+Icons:home (icons)

The assessment tasks are outlined in the next section ->

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1.7. Tasks

To proceed to higher parts, **you must complete** all the specifications in the lower part; you must **not** cherry-pick specifications from various parts. For example, complete all the specifications in the PA part before proceeding to the CR part, and so on.

Using hooks, create a client-*side* prototype for the VV web system with FUNCTIONAL React TS *with the latest version*. Use of object-oriented React / React JS / databases will FETCH A ZERO for the whole assignment.

The tasks are shown below:

PA part [13 marks]

a. (2 marks) **Styling and content of pages**

You should **not** use readymade online templates; you MUST create your own template using CSS or any styling library.

> CSS libraries such as Tailwind and Chakra UI are allowed. Component libraries that provide pre-built navbars and other non-primitive components are not allowed.
>
> The core purpose of this is so that you know how to create these layouts yourself. If you are unsure if a particular library is allowed, please get in touch with your tutor.

Your web application must have clearly distinguishable areas, such as a *header, footer, main content area,*
and/or other *sections, and t*hese areas must be broken into individual components.

The landing page should display information about the VV web system and a navigation bar with appropriate links.

For this part, you will be marked on the styling, layout, and how components are defined. Writing bloated components, ungainly page designs, or not following general web design principles will lead to a loss of marks.

b. (3 marks) **Sign-up** & **Sign-in**

Signup: full implementation is NOT required at this stage. You will complete this for assessment 2. Just leave the text link and do not write any code for the registration feature.

Store some dummy user registration data (email and password) in HTML's local Storage object.

Sign-in: The sign-in component should be a form with only email and password fields. After the proper validations (email and **strong** password) and checks have been performed, the user will be redirected to an appropriate page, as explained in the following few tasks. Provide a visual cue upon successful login. The visual cue could be a text or a pop-up message.

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Add CAPTCHA feature to secure the above sign-in process. *This will require some research on your part.*

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c. (3 marks) **Hirers page**

After the sign-in process, the hiring applicants should be presented with a user interface that allows them to

i. Enter and update hirers profile including name and phone number. Please note email and password was already registered during sign-up.

ii. view a list of pre-populated venues should be made available for the hiring applicant to select as venue candidates.

iii. rank the chosen venue candidates in the order of preference;

iv. apply for a venue to hire for events they are organising which should include extra details such as event name, number of expected guests, and date, time and duration of proposed event;

v. view their hiring reputation, which is based on the historical list of previous venues

that they hired which show details including: venue name and location, event name, date of

hire, and an out-of-5-star rating.

All user inputs used to implement this feature must be validated.

d. (5 marks) **Vendors page**

Vendors should be presented with a user interface that allows them

i. to view the list of applicants based their event requirements and suitability;
ii. to view the historical hire list and compliant documents to check hirer's credibility;
iii. select the appropriate applicants;
iv. make comments for the chosen hiring candidates;
v. approve the application and confirm the booking;

CR part [4 marks]

e. (4 marks) **Additional features for Hirers and Vendors**

Modify the above user interface to add the following features:

i. hirer should be able to search venues based on the venue name, location, capacity, and recommended suitability;
ii. vendor should be able to sort the applicants' list by reputation score. The reputation score is based on a given rating between 0-5 stars by previous vendors. Unhappy vendors would give low score, whereas satisfied vendors would give high score.
iii. A vendor should be able to block any venue for a period, for example during

maintenance / renovation of the venue building and facilities, effectively making it unavailable for hire.

Note: both DI and HD parts require rigour, research, and effort. You will not find direct answers anywhere; you should do some reading on the Internet before devising your own implementations. DO NOT COPY/PASTE solutions from ChatGPT and/or CoPilot. We will conduct a face-to-face interview if your code has been flagged as suspicious.

DI part [3 marks]

f. (3 marks) **Visual representation**

Vendors should be able to see an appropriate visual representation of the following data:

i. most chosen applicant
ii. least chosen applicant
iii. the applicants who have not been selected

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Make use of a custom hook, to program all of the features in DI part.

HD part [5 marks]

(3 marks) **Advanced requirements (requiring your own research)**

The hiring applicants must now submit compliant documents which will give them stronger credibility. These documents should be scanned / photographed and saved in various file formats:

- Applicant's driver's license (jpg)
- Public liability insurance certificate of currency document (pdf)
- Certificate of registration for business name (pdf) if the applicant is applying for venue on behalf of a business / organisation. You need to have a way to check if they are part of a business, e.g. a tick box that will show a section for entering ABN Number and the field to upload the pdf file.

Technical note: At this stage you must store the files in HTML5's localStorage.

Credibility score between 0-5 stars is calculated automatically by the system between 0-5 stars and relative to the number of compliant documents submitted. For example, if hirer did not submit any ID, this will be counted as 0 star. The stronger their document evidence they submit, the more stars they will get.

(2 marks) **Unit Testing**

g. Write three unit tests in the context of the features covered above.

Good unit tests for a front-end React web application should focus on verifying that components render correctly, handle state and props properly, and respond to user interactions (like clicks or form submissions) as expected. They should ensure proper handling of conditional rendering, edge cases, as well as test the behaviour of hooks like useEffect. Tests should check that components manage form input and validation correctly, handle errors gracefully, and ensure accessibility features (like semantic HTML and keyboard navigation) are in place.

NOTE: simplistic implementations for DI and HD parts which do not implement above will receive very low marks.

1.8. Marking Guidelines

The marks allocated have been added to each of the tasks. **However, you will need to read the rubric for details**.

All the best!